UNIT 1
BUILDINGS, BUILDING MATERIALS

Definition of Building
National Building Code of India has defined building as 'any structure constructed for any purpose with any materials, used for human habitation or not'.
It includes foundation, plinth walls, floors, walls, roofs, chimneys, fixed platform, verandah, balconies, plumbing and building services, etc.

Classification of the Buildings according to the National Building Code (NBC):

National Building Code of India has classified buildings based on occupancy and types of construction.

1. Residential Buildings: Sleeping accommodation for normal residential purpose is provided with or without cooking or dining or both.
Example: houses, apartments, hostels, lodge, etc.

2. Educational Buildings: Assembly facilities for instruction or education are provided.
Example: schools, colleges, etc.

3. Institutional Buildings: Medical or other treatment or special care is provided.
Example: hospital, asylums, prisons, old age homes, school for special children, etc.

4. Assembly Buildings: Gathering of people for amusement, recreation, social, religious, civil, travel, etc.
Example: theatre, cinema halls, assembly houses, auditorium, museum, restaurants, places of worship, terminals for road, rail, air and sea transport (railway station, airports, etc), play grounds, stadiums, etc.

5. Business Buildings: Business transactions, keeping of records, service facilities are provided. **Example:** court houses, libraries, offices, banks, etc.

6. Mercantile Buildings: Display and sale of merchandise - either wholesale or retail takes place. **Examples:** shops, stores, markets, show rooms, etc.

7. Industrial Buildings: Products or materials are fabricated, assembled, processed or manufactured.
Example: factories, laboratories, power plants, refineries, dairies, saw mills, assembly plants, etc.

8. Storage Buildings: Storage or sheltering of goods, vehicles, animals, wares or merchandise except which are combustible or explosive material.
Example: warehouse, cold storage house, truck and marine terminal garages, hangar, grain silos, barns, stable, etc.

9. Hazardous Buildings: Storage, handling, manufacture or processing of highly combustible or explosive materials.
Example: hazardous chemical factories, explosive factories and godowns, arms and ammunition factories for army, etc

Classification based on type of construction: considered by the fire resistance of the building.
Type-1: That provides 4 hours fire resistance
Type-2: That provides 3 hours fire resistance
Type-3: That provides 2 hours fire resistance
Type-4: That provides 1 hour fire resistance

Definitions of Areas of building:

PLINTH AREA
Plinth area is defined as the total covered built-up area of a building, measured at floor level or any storey.
It is calculated by measuring the external dimensions of a building. It includes supported porches (non-cantilever) and excludes courtyards, open areas, balconies, and cantilever projections.

FLOOR AREA
Floor Area of any floor is the total floor area between walls for all rooms in that floor.
It includes area of verandas, passages, corridors, stair rooms, entrance halls, kitchens, stores and toilets. It excludes area of all walls, columns / pillars and intermediate supports.

Floor Area = Plinth Area - Area of all walls.

CARPET AREA
Carpet area of any floor is the useful or livable area of that floor. Or it is the area where floor carpet is laid. It excludes toilets, passages, verandahs, corridors, stairs, lifts, entrance hall, etc.

CIRCULATION AREA
Circulation Area is defined as the area of building that is used for the movement of people inside the building.
It includes verandahs, passages, corridors, balconies, entrance halls, porches, stairs, etc.

FLOOR AREA RATIO
Ratio of total built-up covered area to the area of the plot.

$$\textbf{Floor Area Ratio (FAR)} = \frac{\textbf{Total plinth area of all floors}}{\textbf{Plot area}}$$

BUILDING MATERIAL

Building material or construction material is the material used for building or constructing the structures like building, dams, roads, etc.

Example:
Solid materials like stone, sand, timber, steel
Binding materials like cement, lime, mortar
Protective materials like paint, varnish, plaster

STONE

TYPES OF STONES

Based on their **origin of formation,** rocks are classified into three main groups

- Igneous rocks. Eg: Granite and basalt
- Sedimentary rocks. Eg. Sand stone and lime stone
- Metamorphic rocks. Eg: Marble and slate

Based on their **Physical appearance**,

- Stratified rock (eg: limestone)
- Un stratified rock (eg: granite)
- Foliated rock (eg: gnesis)

Based on their **constituents**,

- Siliceous rock (eg: granite)
- Calcareous rock(eg: limestone)
- Argilaceous rock (eg:slate)

Quarrying: It is the process by which stones are obtained from rock. It is quarried by hand tools, by blasting and using some machinery.

Dressing: The process of bringing the stones to a regular shape and required surface finish is known as dressing.
Types: Quarry dressing and site dressing

QUALITIES (REQUIREMENTS) OF A GOOD BUILDING STONE

- It should be hard and its crushing strength value should be greater than 100 N/mm^2
- It should not absorb water more than 0.6 % by weight
- Its specific gravity value should be greater than 2.7
- Its color should be uniform & not affected by weathering
- It should be durable and it should have compact and fine crystalline structure
- It should have acid resistance and it should be free from soluble matter
- It should be easily dress-able, i.e. it can be easily or with less effort, moulded, cut, carved and smoothened
- Its surface should be free from cavities & cracks
- Its fracture should be sharp, even and clear
- It should not be easily affected by fire
- Its surface should be dry (free from moisture)

USES OF STONE

- Stones are used in construction of walls, columns, lintels, roofs , flooring, arches, etc
- Stone blocks are used in construction of bridge piers and abutments.
- Stone blocks are used in construction of big dams.
- Stone panels are used for face works of buildings
- Stone blocks are used in paving of roads and footpaths
- Stone aggregates are used as railway ballast
- Stones are used as basic inert material in concrete.
- Stones are used as flux in blast furnace

TESTS CONDUCTED ON STONES:

1. **Crushing Strength Test:**
   This test is conducted to know the static load at which the stone fails or is crushed. For conducting this test, specimen is prepared from parent stone and tested under the compression testing machine. The crushing load is noted. Then crushing strength is equal to the crushing load divided by the area of specimen over which the load is applied.

2. **Water Absorption Test**:
   This test is conducted to know the amount of water absorbed by the stone. The specimen is weighed when it is completely dry and free from moisture. Then it is immersed in water for 24 hours and surface cleaned and the wet weight is taken. The difference between these 2 readings gives the weight of water absorbed by stone.

3. **Abrasion Test:**
   This test is carried out on stones which are used as aggregates for road construction. The test result indicates the suitability of stones against the grinding action under traffic. Any one of Los Angeles abrasion test, Deval's abrasion test or Dorry's abrasion test may be conducted to find out the suitability of aggregates.

   Los Angeles abrasion test is commonly followed as it gives near accurate results. Steel balls of specified dimensions are mixed with test specimen in a big motorized steel drum which recreates the effect of grinding effect by traffic. The abrasion value is the percentage of fines. For bituminous mixes recommended value is 30% and for base course it is 50%.

4. **Impact Test:**
   Impact strength of stones means their resistance to falling load. It is found by using impact testing machine. A specified weight is made fall freely over the given specimen for specified times. The fines are weighed and impact strength of total sample is expressed as the percent of fines and it is known as impact value. For wearing course recommended value of impact value is 30%, for bituminous macadam 35% and for water bound macadam 40%.

5. **Acid Test:**
   This test is normally carried out on sand stones to check the presence of calcium carbonate, which weakens the weather resisting quality. In this test, a sample of stone is kept in a solution of one per cent hydrochloric acid for seven days. A good building stone maintains its sharp edges and keeps its surface intact. If edges are broken and powder is formed on the surface, it indicates the presence of calcium carbonate. Such stones will have poor weather resistance.

BRICK

Brick is a building block like stone and after stone it is the oldest building block being used. Brick is obtained by moulding good clay into a block of required dimensions, which is dried and then burnt.

CONSTITUENTS OF BRICKS

Silica : It forms **50 % - 60 %** of bricks and gives shape, prevents cracking, shrinkage
Alumina : It constitutes **20 % - 30 %** and gives plasticity to bricks
Lime : It is added **up to 5%** and used to prevents shrinkage and melts bricks
Iron oxide: It is added from **5 % - 6%** to give red colour to bricks
Magnesia : It is added in **small quantity** to give yellow tint and to decrease shrinkage

MANUFACTURE OF BRICKS

The steps followed in the manufacture of the burnt clay bricks are

1) Preparation of earth:

It consists of the following steps,

- Removal of loose soil of about 20 cm is done as it contains lot of impurities
- Digging, spreading and cleaning of the soil is done to remove all undesirable matters like stones, vegetable matter, etc.
- Weathering of soil is permitted for weeks to full season for its softening
- Blending of soil with its ingredients is done by taking a small portion of clay and turning it up and down in vertical direction.
- Tempering is done by adding water to soil-clay mixture and kneading by men or cattle or pug mill to make it homogeneous and plastic.

2) Moulding: The tempered mixture is moulded into wooden or steel moulds of required dimensions over ground or table either by men or machines. Table moulded bricks have plane surfaces than ground moulded bricks. Machine moulded bricks have smooth surfaces, sharp edges and corners and hence are of high quality.

3) Drying: Bricks are dried for about 5 to 12 days in drying yards before burning, else they will wrap due to sudden evaporation of moisture.

4) Burning: Dried bricks are burnt to impart strength into them. Since both under burnt and over burnt bricks don't posses required strength, bricks need to be burnt for specified time only. Burning of bricks is done in kilns or clamps.

CLASSFICATION OF BRICKS

First Class: They are table moulded, with uniform crimson red colour and size, perfect edges and are properly burnt. They are used for superior works.

Second Class: They are ground moulded, with slightly dark red colour, non perfect edges and are slightly over burnt. They are used where brick masonry is plastered.

Third Class: They are ground moulded, with light colour, uneven edges and surface and are under burnt. Used for unimportant and temporary constructions.

Fourth Class: They are ground moulded, with dark red colour, broken edges and surface and are over burnt. Used for concrete foundations, floors, roads, etc.

TYPES OF BRICKS

- Building Bricks
- Paving Bricks and Fire Bricks
- Special Bricks like specially shaped bricks, facing bricks, perforated building bricks, burnt clay hollow bricks, sewer bricks and acid resistant bricks.

PROPERTIES / REQUIREMENTS OF A GOOD QUALITY BRICK

- Color should be uniform and bright copper colour or brick red colour
- Size should be standard and uniform
- They should have shape with flat surface faces and perfect right angled edges
- They should possess fine, dense and uniform texture and they should not possess fissures, voids, loose grit and un-burnt lime.
- When struck with each other clear ringing sound should be produced
- Water absorption should not be greater than <u>20 %</u> when immersed in water for 24 hours
- Weight of each brick should be from <u>3 to 3.5 kg</u>
- Crushing strength should be <u>more than 3.5 N/mm^2</u>
- Thermal conductivity should be less and should be soundproof
- <u>No efflorescence</u> (deposits of salts when immersed in water) should appear
- They should posses hard, cracks-free surface and show no impression when scratched with nail.
- Should not break when dropped flat from <u>height of 1 m</u>.
- Fire resistance of bricks is usually high.

USES OF BRICKS

- Bricks are used in masonry as building blocks to be used in walls, columns, foundations, arches, etc.
- They are used to build dams, bridges, etc.
- They are used as building blocks in laying drains, sewers and floor.
- They are used as refractory material in constructing ovens, furnace and chimneys, etc.
- High quality bricks are used for ornamental works
- First quality bricks are used for facing of walls without plastering
- Bricks are used to protect steel columns from fire.
- Brick bats or broken pieces of bricks are used as aggregates in concrete

TESTS ON BRICKS

The following ***laboratory tests*** may be conducted on the bricks to find their suitability:

(*i*) **Crushing Strength**: The specimen is placed in compression testing machine load is applied axially and the crushing load is noted. Then the crushing strength is the ratio of crushing load to the area of brick loaded. Average of five specimens is taken as the crushing strength.

(*ii*) **Water Absorption Test**: Brick specimen are weighed dry. Then they are immersed in water for a period of 24 hours. The specimen are taken out and wiped with cloth weight is determined. The difference in weight indicates the water absorbed. The average of five specimens is taken. This value should not exceed 20 per cent.

(*iii*) **Shape and Size**: Bricks should be of standard size of 190 mm × 90 mm × 90 mm and edges should be truly rectangular with sharp edges. To check it, 20 bricks are selected at random.

(*iv*) **Structure:** A few bricks may be broken in the field and their cross-section observed. The section should be homogeneous, compact and free from defects such as holes and lumps.

(v) **Sound Test:** If two bricks are struck with each other they should produce clear ringing sound. The sound should not be dull.

(*vi*) **Hardness Test**: For this a simple field test is scratch the brick with nail. If no impression is marked on the surface, the brick is sufficiently hard (*vii*) **Efflorescence:** The presence of alkalis in brick is not desirable because they form patches of gray powder by absorbing moisture. Hence to determine the presence of alkalis this test is performed.

ADVANTAGES OF BRICKS

- Bricks can be easily manufactured from locally available cheap raw materials
- The manufacturing process of bricks can be mechanised and hence more labour is not required
- The bricks can be manufactured of required uniform shape and size
- Bricks have less unit weight than stone and hence it has less self weight
- Due to their lesser weight than stone, bricks are easier to handle and cheaper to transport
- Bricks are more sound proof than stone

DISADVANTAGES OF BRICKS

- Manufacture of bricks require burning which consumes lot of fuel and causes air pollution
- Bricks cannot be dried during rainy season and hence manufacturing process is hindered
- Bricks are more brittle than stone and their edges gets damaged easily during handling and transporting
- Bricks are less harder and less durable than stone
- Generally brick masonry requires plastering as they are less resistant to weathering and absorb water
- Its life is less and hence maintenance cost is more

CEMENT

Cement is a commonly used binding material in construction. It is in powdery form and mixed with water to undergo hydration to form a hard and strong building material. It can be used in mortar as a covering and bonding material and in concrete as building material.

CONSTITUENTS OF CEMENT

The Basic constituents of cement are

Lime : **62 %** is added to cement to give its binding property
Silica : **22 %** is added to give strength to cement
Alumina : **5%** is added to help in hydration of cement
Calcium sulphate: **4%** retards setting time of cement gypsum
Iron oxide : **3%** is added to give colour and to help in hydration of cement
Magnesia : **2%** is added to help in hydration of cement
Sulphur : **1%** is present as impurity
Alkalis : **1%** is added to help in hydration of cement

PRODUCTS OF HYDRATION OF CEMENT OR SETTING OF CEMENT

Tri-calcium silicate $3CaO\ SiO_2$ (C_3S)
Di-calcium silicate $2CaO\ SiO_2$ (C_2S)
Tri-calcium aluminate $3CaO\ Al_2O_3$ (C_3A)
Tetra-calcium aluminate $4CaO\ Al_2O_3\ Fe_2O_3$ (C_3AF)

MANUFACTUIRNG OF CEMENT

(i) Mixing of raw materials
(ii) Burning
(iii)Grinding

Mixing of raw materials : There are two methods of mixing of raw materials (Dry process and wet process). In dry process the calcareous and argillaceous materials are reduced to proper sizes, ground in ball or tube mills, mixed in correct proportions and stored in tanks.

In wet process, the calcareous materials are crushed to required size and then argillaceous materials are washed and stored in basins. Both are mixed in a required proportion in a channel, ground in ball or tube mill to form thick liquid called slurry which is stored in tank.

Burning: The dry mixture of the slurry and burnt in rotary kiln at 1500 to 1700°C. The materials chemically get fused to form hard balls of cement called clinkers which collected in containers.

Grinding: The clinkers are ground with 3 to 4% of gypsum in a ball or tube mill. Gypsum is added to control the initial setting time. The finely ground cement is stored in silos. Cement is then packed in bags containing 50 kg. Such bags are stored in dry place before dispatch.

TYPES OF CEMENT AND THEIR USES

Ordinary Portland Cement (OPC)	is used in all common cement works as mortar and concrete
Portland Pozzolanic Cement (PPC)	is used where sulphate resistance is required
White Cement	is used for ornamental purposes and for fixing marbles and glazed tiles
Coloured Cement	is used for finishing touches to floors, walls, etc.
Expanding Cement	is used in repair works
Low Heat Cement	is used in construction of dams and other massive structures
Quick Setting Cement	is used in under water construction
Rapid Hardening Cement	is used in rapid construction works
Acid Resistant Cement	is used in flooring of chemical factories
Sulphate Resisting Cement	is used in lining canals, culverts, etc.
Blast Furnace Cement	Is used for lining chimneys, furnaces, etc

GRADE OF CEMENT

The ordinary Portland cement is represented by its characteristic compressive strength at 28 days, expressed in N/mm^2. The types of grades of cement are **33, 43 and 53**.

PROPERTIES OF CEMENT

- Its color should be uniform
- It should be dry fine powder without lumps
- Its specific gravity should be 3.15 and it should sink in water
- It should be cool and uniform when touched
- It should have fineness of 2250 mm^2/gm of specific surface
- Its initial setting time shall not be less than 30 minutes and its final setting time shall not be greater than 10 hours.

- When it undergoes Soundness test its expansion shall <u>not be greater than 10 mm.</u>
- It should be easily workable and when mixed with water and should posses good plasticity and binding property

USES OF CEMENT

- It is the extensively used building material in construction as binding material in mortar, concrete, etc
- It is used in mortar for wall masonry and plastering of walls, ceilings, etc. and also in floor finishing
- It is part of concrete which is widely used construction material for constructing buildings, dams, bridges, etc.
- Concrete with steel as RCC are used in foundation, column, wall, beam, roof, floor, etc.
- White & coloured cement are used for decorating works like wall plastering
- Cement is used to manufacture cement sheets / panels to be used in roofs, walls, etc.
- Cement is used to make pipes for transporting water and other liquids.
- Cement is used to make tiles for using in floors, roofs, etc

MORTAR

Mortar is a mixture of a binding material like cement, water and fine aggregate like sand. It forms a wet paste while mixing and becomes strong after setting. Cement mortar and lime mortar are examples for mortar.

CEMENT MORTAR

Cement mortar is a mixture of a binding material (cement), water and the fine aggregate sand. It forms a wet paste while mixing and becomes strong after setting. It is used for plastering of walls and for bonding bricks in wall masonry.

INGREDIENTS OF MORTAR

- Binding material (like lime or cement)
- Fine aggregate (like sand or crushed rock)
- Water

GRADES OF CEMENT MORTAR

Cement mortars are assigned certain grades based on their minimum compressive strength at 28 days curing and mix proportion by volume.

GRADE	MORTAR MIX (CEMENT: SAND)	COMPRESSIVE STRENGTH AT 28 DAYS	USES
MM 1.5	1:7	1.5 to 2 N/mm^2	Foundation
MM 3	1:6	3 to 5 N/mm^2	Masonry
MM 5	1:5	6 to 7.5 N/mm^2	Plastering
MM 7.5	1:4 to 1:3	7.5 N/mm^2 and above	Floor finishing and damp courses

PROPERTIES OF MORTAR

- It should be cheap and durable
- It should be easily workable
- It should have good binding with brick, stone, etc.
- It should not crack in joints
- It should set and harden quickly
- It should have resistance to rain water after setting
- It should have required strength
- It should be tough and durable after hardening
- It should have resistance to fire and weathering

USES OF MORTAR

Cement mortar is used as bonding material in masonry to bond bricks or stones

- It is used for plastering walls and ceilings and finishing floors and slabs, etc.
- It is used as finishing material in all concrete works
- It is used in manufacturing building blocks
- It is used as filler material in stone masonry
- It is used for pointing of masonry work
- It is used in repair works
- It is used as wearing course for roads, floors, etc.

CONCRETE

Concrete is a mixture of binding material, fine aggregate, coarse aggregate and water. It can be easily moulded to desired shape when it is wet and workable. After setting of the binding material (cement) it hardens like stone.

CONSTITUENTS OF CONCRETE

- Binding material (like lime or cement)
- Water
- Aggregate:
  i. Fine aggregate (like sand or crushed rock) – 4.75 to 0.07 mm
  ii. Coarse aggregate (like stone or brick bats) – 75 to 4.75mm

CONCRETE GRADE

The concrete mix is represented by its characteristic compressive strength at 28 days expressed in N/mm^2. This is denoted by adding the prefix M such as M15, M20, etc.

GRADE	MORTAR MIX (CEMENT: SAND:COARSE AGGREGATE)	COMPRESSIVE STRENGTH AT 28 DAYS	USES
M 10	1:3:6	10 N/mm^2	Mass concrete in superstructure (piers and abutments)
M 15	1:2:4	15 N/mm^2	All normal RCC works (beams, columns, etc)
M 20	1:1.5:3	20 N/mm^2	Water retaining structures
M 25	1:1:2	25 N/mm^2	High load columns and beams

FACTORS AFFECTING COMPRESSIVE STRENGTH OF CONCRETE

The various factors affecting the compressive strength of concrete are:

- Cement content
- Strength of fine aggregates and coarse aggregates
- Water cement ratio
- Quality of water used
- Presence of air voids
- Admixtures
- Compaction of concrete
- Period of curing
- Age of concrete

TYPES OF CONCRETE

- Plain Cement Concrete (PCC)
- Reinforced Cement Concrete (RCC)
- Fibre Reinforced Concrete (FRC)
- Pre-stressed Concrete (PSC)
- Light Weight Concrete
- Aerated Concrete
- High Density Concrete
- No-fines Concrete

PROPERTIES OF CONCRETE

- Concrete is the widely used construction material in Earth, almost no structure is built without concrete
- Concrete is like plastic and workable when fresh i.e. when wet. It can be moulded to any shape based on form work.
- Concrete can be manufactured of any desired strength by proportioning its ingredients.
- It shrinks during hardening due to loss of water and hence curing is done. It also shrinks with age, but to a negligible extent.
- Concrete is durable, fire resistance and water proof, once it has hardened
- It is economical, its strength to cost ratio is high.
- It is strong in compression but weak in tension. By using steel reinforcements, (RCC) can withstand both tension and compression.
- It is maintenance free and no strength is lost with age
- It is free from corrosion.
- It should sustain all weathering conditions.

USES OF CONCRETE

- Bed concrete is used below column footings, wall footings and on wall at supports to beams
- In pavements and for making building blocks
- RCC is used in structural element like footings, columns, beams, lintels, roof slabs
- It is used for the construction of storage structures (water tanks, dams, silos and bunkers), big structures (bridges, retaining walls, docks, harbours and under water structures) and tall structures (multi storey buildings, chimneys and towers)
- It is used for pre-casting in making railway sleepers and electric poles

MANUFACTURING OF CONCRETE

Batching is the measurement of materials for making concrete like cement, water, sand and stones. Concrete can be weight batched or volume batched. In volume batching, cement, sand and stones are batched by their volume with help of wooden box or metal container. It gives varying degree of mix proportions as the stones and sand have varying volume. In weight batching, cement, sand and stones are batched by their weight. It gives accurate mix proportions and hence high quality concrete.

Mixing is the process of blending the constituents of concrete together either by hand or machine. First cement, sand and stones are blended to form uniform color and then water is added to it. This mixture is mixed thoroughly such that the water mixes with cement and should not result in separation of the constituents.

Transporting of the above mixture is done by manual, wheel burrows, conveyor belts, chutes, mixer trucks, etc. to the place where it has to be placed.

Placing of the above mixture is done manually or by chutes or pipes in to its form works and it is ensured that it is not dropped.

Compacting is done to expel the entrapped air voids either by hand or vibrator. Vibrators are may be external type like form works or internal type like needle or plate vibrators. Excess vibration as well as less vibrations result in loss of concrete’s strength.

Curing is the process of adding water to maintain satisfactory moisture and temperature conditions for setting of cement. During the setting of cement, the heat evolved due to hydration of cement (exothermic reaction) causes the evaporation of water which is present for further reaction. To compensate this water loss, water is added frequently during the setting period of cement. If curing water is not added setting is not completed and cement will not attain its required strength. It can be done by spraying of water, covering the surface with wet gunny bags, straw etc, by ponding, steam curing and application of curing compounds.

MAIN PURPOSE OF USING REINFORCEMENT IN CONCRETE

Concrete is good in compression but weak in tension. Hence some materials like steel which have good tensile strength are used as reinforcement in concrete and are embedded inside the concrete. RCC is very common building material and is used in all parts of a building.

STEEL

Steel is an alloy of iron and carbon (0.25 % to 1.5 %). It is the most common metal as well as high strength material used as building material.

TYPES OF STEEL

- Mild steel or Low carbon steel - carbon content up to 0.25 %
- Medium carbon steel - carbon content from 0.25 % to 0.70 %
- High carbon steel and - carbon content from 0.70 % to 1.5%

PROPERTIES OF STEEL

- It has high tensile strength and compressive strength.
- It is can be welded and forged

- It is ductile and malleable
- Its specific gravity is 7.85 and its melting point is around 1400^{o}C
- Its Young's Modulus is 2.10 x 10^5 N/mm^2
- It can be magnetised permanently

USES OF STEEL

- Various sections of steel bars are used as reinforcement in RCC works, as columns, beams, trusses etc
- Steel sheets are used as roofing materials and as walls for water tanks, etc.
- Steel is used in making parts of various machines and for making all kinds of tools

ADVANTAGES AND DISADVANTAGES OF STEEL

Advantages of steel:

- It has good tension as well as compression strength.
- Its strength to mass ratio is high than the other construction materials and the structures are light weight.
- Its cost to strength ratio is higher.

Disadvantages of steel:

- The major disadvantage of steel is that it undergoes rusting or corrosion.
- It is due to the oxidation of steel.
- It can be prevented by avoiding the direct exposure of the steel to atmosphere, by coating it with paints or protective chemicals or by galvanisation. Hence maintenance of steel is costlier.

PROPERTIES OF MILD STEEL

- It is ductile, malleable and more elastic
- It has fibrous structure and can be magnetized permanently
- Its specific gravity is 7.8 and its Young's modulus is 2.1 × 10^5 N/mm^2
- It can be forged and welded
- It is equally strong in tension and in compression
- Its melting point > 1400^{o}C

USES OF MILD STEEL

- It is used as reinforcing bars in RCC
- As angles & other sections, bars or rods, sheets or plates, it is used in construction of structural members
- Used in machine parts, motor body, tubes, wires, sheet plate, tin metal, etc.
- It is also used for manufacturing rails, cranes, towers, etc.

PROPERTIES OF MEDIUM CARBON STEEL

- Its strength is more than that of mild steel
- It cannot be easily forged and welded
- It can withstand shocks and vibrations
- It can be hardened to little extent
- It is tougher and harder than mild steel

USES OF MEDIUM CARBON STEEL

- It is used as structural steel as various sections (rod, bar, flat, channels, 'I' and 'L' angles and 'T' sections)
- It is also used to make boiler plates, rails, hammers, pressing dies, etc

PROPERTIES OF HIGH CARBON STEEL

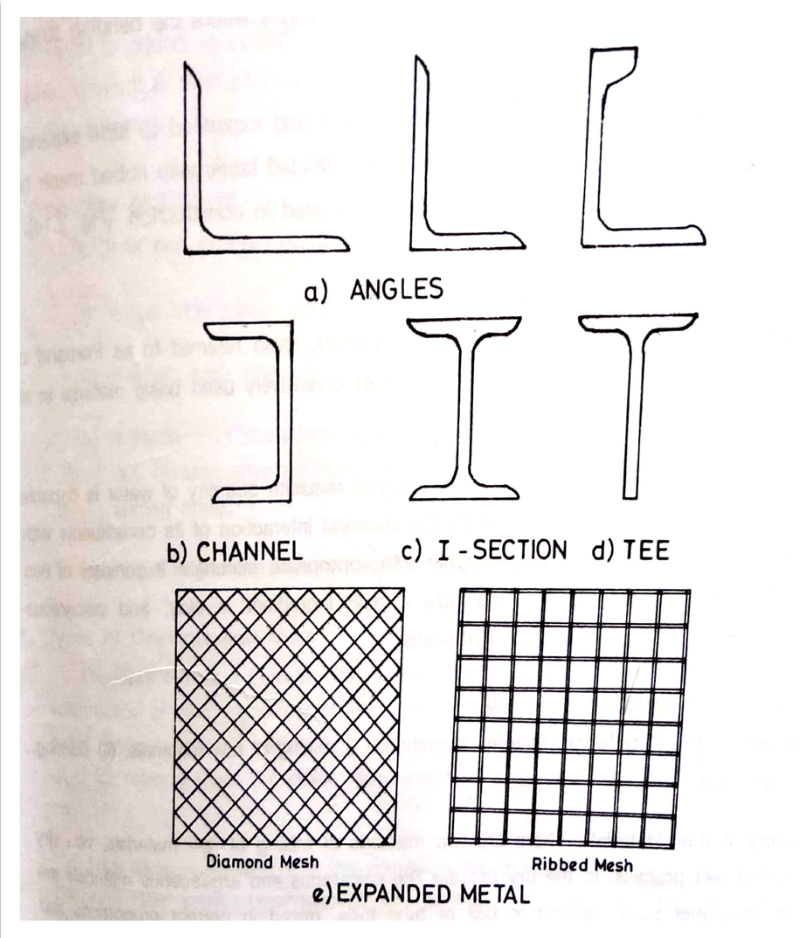
- It has granular structure and can be magnetised permanently
- It is readily forged and welded
- It is tougher than mild steel and it can withstand shocks and vibrations
- It can be easily hardened and tempered
- Its specific gravity ~ 7.9 and melting point is > 1300^{o}C

USES OF HIGH CARBON STEEL

It is used to make sledge hammers, springs, smith's tools, metal cutting tools, planning machines, wood working tools, axes & hammers, mining equipments, drill bits, stone masons' tools and knives & forks, etc.

VARIOUS FORMS OF REINFORCING STEEL

Mild steel round rods, tor steel rods, steel strands or wires, flat bars, special sections like 'L', 'I', 'C', 'T', etc.



COMPARE ORDINARY STEEL WITH MILD STEEL

Unlike ordinary mild steel bars which has plain surface, tor steel has ribbed perimeter and it is used in RCC as reinforcement. Its area of contact with concrete is more when compared to mild steel rod of same diameter. Therefore its bonding strength with concrete is higher than that of mild steel rods.

Tor steel is also tempered while manufacturing and hence it has high strength than mild steel.

COMPARE THE PROPERTIES OF THREE TYPES OF STEEL

Property	**Mild steel**	**Medium carbon steel**	**High carbon steel**
Carbon content	0.10 % to 0.25 %	0.25 % to 0.60 %	0.60 % to 1.50 %
Molecular structure	Fibrous structure	Fibrous structure	Granular structure
Ductility	More ductile	Medium ductile	Less ductile
Specific gravity	7.8	7.8 - 7.9	7.9
Weld-ability & forge-ability	Easily done	Cannot be done	Easily done
Magnetisation	Easily magnetised	Cannot be magnetised	Easily magnetised
Melting point	> 1400°C	> 1350 °C	> 1300°C
Strength	High	Higher than mild steel	Highest
Hardening & tempering	Not possible	To little extent	Can be done
Toughness & hardness	High	Higher than mild steel	Higher than mild steel
Shocks & vibrations	Cannot withstand	Can withstand	Can withstand

TWO MARKS- UNIVERSITY QUESTIONS WITH ANSWERS

1. **Distinguish plinth area and carpet area of a building.(jan 2013,may 2013) or Define plinth area. (Nov 2018) or Define carpet area.(jan 2016)**

   Plinth Area is defined as the total covered built-up area of a building, measured at floor level or any storey. It is calculated by measuring the external dimensions of a building. It includes supported porches (non-cantilever) and excludes courtyards, open areas, balconies, and cantilever projections.
   Carpet Area of any floor is the useful or livable area of that floor. Or it is the area where floor carpet is laid. It excludes toilets, passages, verandahs, corridors, stairs, lifts, entrance hall, etc.

2. **What do you mean by grade of cement? (jan 2010)**

   The ordinary Portland cement is represented by its characteristic compressive strength at 28 days, expressed in N/mm^2. The types of grades of cement are 33, 43 and 53.

3. **What do you mean by floor area? (april 2010)**

   Floor Area of any floor is the total floor area between walls for all rooms in that floor. It includes area of verandas, passages, corridors, stair rooms, entrance halls, kitchens, stores and toilets. It excludes area of all walls, columns / pillars and intermediate supports.
   Mathematically,
   Floor Area = Plinth Area - Area of all walls.

4. **List the types of concrete? (april 2010)**

    - Plain Cement Concrete (PCC)
    - Reinforced Cement Concrete (RCC)
    - Fibre Reinforced Concrete (FRC)
    - Pre stressed Concrete (PSC)
    - Light Weight Concrete
    - High Density Concrete

5. **What are the various form of reinforcing steel. (nov 2011)**

    - Mild steel round rods
    - Tor steel rods
    - Steel strands or wires
    - Flat bars
    - Special sections like 'L', 'I', 'C', 'T', etc.

6. **Define engineering and civil engineering. (jan 2010)**

    - Engineering is the field of application of the various scientific principles for the benefaction of mankind.
    - Civil Engineering is oldest branch of engineering, directing the great resources of Nature for the use and convenience of mankind.

7. **List the tests conducted on stones. (jan 2011)**

- Crushing strength test
- Water absorption test
- Abrasion test
- Impact test
- Acid test

8. **What are the qualities of good brick? (april 2011,may 2016)**

- Uniformity in colour, shape, and size
- Sufficiency in soundness, hardness and strength
- Nil efflorescence, less water absorption and low thermal conductivity
- Good sound insulation and fire resistance

9. **Explain the uses of tor steel. (april 2011) or Compare ordinary steel with tor steel. (jan 2013)**

    - Unlike ordinary mild steel bars which has plain surface, tor steel has ribbed perimeter and it is used in RCC as reinforcement. Its area of contact with concrete is more when compared to mild steel rod of same diameter. Therefore its bonding strength with concrete is higher than that of mild steel rods.
    - Tor steel is also tempered while manufacturing and hence it has high strength than mild steel.

10. State some types of cement. [May, 2009]

Ordinary Portland Cement (OPC), Portland Pozzalona Cement (PPC), White Cement, Coloured Cement, Quick Setting Cement, Rapid Hardening Cement, Expanding Cement, Acid Resistant Cement, etc.

11. Define cement mortar. (nov 2011,april 2012)

Cement mortar is an example of mortar and it consists of cement as binding material, water and the fine aggregate sand. It forms a wet paste while mixing and becomes strong after setting. It is used for plastering of walls and for bonding bricks in wall masonry.

12. List out different types of bricks (april 2011)

- Building Bricks
- Paving Bricks and Fire Bricks
- Special Bricks like specially shaped bricks, facing bricks, perforated building bricks, burnt clay hollow bricks, sewer bricks and acid resistant bricks

13. Why reinforcement is used in concrete? [May, 2009] or What do you mean by reinforced concrete? (jan 2013)

Concrete is good in compression but weak in tension. Hence some materials like steel which have good tensile strength are used as reinforcement in concrete and are embedded inside the concrete. RCC is very common building material and is used in all parts of a building.

14. What are the raw materials used in the manufacture of cement.(may 2013)

Basic constituents of cement are lime (62%) and silica (22%). Other constituents are alumina (5%), calcium sulphate (4%), iron oxide (3%), magnesia (2%), sulphur (1%) and alkalis (1%).

15. Classify the buildings according to NBC.(nov 2016,jan 2015) or How can you classify building.(Jan 2011)

National Building Code of India has classified building as Residential Buildings, Educational Buildings, Institutional Buildings, Assembly Buildings, Business Buildings, Mercantile Buildings, Industrial Buildings, Storage Buildings and Hazardous Buildings.

16. State how bricks are classified.(nov 2016)

- First class bricks
- second class bricks
- third class bricks
- fourth class bricks.

17. General properties of cement.(april/may 2014, May 2019)

- Its colour should be uniform
- It should be dry fine powder without lumps

- Its specific gravity should be 3.15 and it should sink in water
- It should be cool and uniform when touched
- Its initial setting time shall not be less than 30 minutes
- Its final setting time shall not be greater than 10 hours

18. What are the uses of bricks?(jan 2015)

- As building block
- For lining of ovens, furnaces and chimneys
- As aggregates in concrete
- For protecting steel columns from fire
- As pavers in footpaths and cycle tracks
- For lining sewer lines

19. List the four operations involved in manufacturing of a brick.(april/may 2016 , May 2019)

- Preparation of brick earth
- Moulding of bricks
- Drying of bricks and
- Burning of bricks

20. How concrete is designated by grades?(may 2016)

The concrete mix is represented by its characteristic compressive strength at 28 days expressed in N/mm^2. This is denoted by adding the prefix M such as M15, M20, etc.

21. What are the constituent material of bricks?(nov 2017)

- Silica: It forms 50 % - 60 % of bricks and gives shape, prevents cracking, shrinkage and wrapping
- Alumina : It constitutes 20 % - 30 % and gives plasticity to bricks
- Lime : It is added up to 5% and used to prevents shrinkage and melts bricks
- Iron oxide: It is added from 5 % - 6% to give red colour to bricks
- Magnesia : It is added in small quantity to give yellow tint and to decrease shrinkage

22. What is quarrying and dressing of stones?(nov 2017)

Quarrying is the process by which stones are obtained from rock. The stones are quarried by hand tools, by blasting and using some machinery.

Dressing is the process of bringing the stones to a regular shape and required surface finish.

<u>DETAILED QUESTIONS</u>

1) Explain the terms plinth area , floor area , carpet area and circulation area, floor space index .(jan 2011,nov 2016)

2) **(a) List the characteristics of good bricks. also give its applications. (may 2014 & Nov 2017, May 2019)**

   (b) What are the advantages of using TMT steel rods? (Nov 2018)

3) What is the composition of brick? Explain their types. (jan 2016)

4) Discuss the different test conducted on bricks. State the uses of bricks.(may 2013)
5) What are the requirements of stone which is to be used as a building material(may 2011,jan 2015)
6) What are the various test conducted on building stones. Explain any four. (april 2012)
7) State different types of stones and their uses in construction.(jan 2013, may 2016 & nov 2017)
8) **What are the different types of cement? Explain their properties and uses.(may 2016) explain the properties and uses of cement concrete.(nov 2016)**
9) Explain the constituents of cement and their functions. also its different tests performed on cement. (april 2014,may 2014, jan 2016)
10) Explain various stages of manufacturing cement. (april 2010)
11) What is cement mortar? What are the uses of cement mortar. (may 2013,jan 2013)
12) Explain the various factors affecting the compressive strength of concrete.(april 2011)
13) Explain various stages of manufacturing concrete. (april 2010)
14) Write the influence of the ratio of the ingredients of concrete. (Nov 2018)
15) What are the different types of steel? Explain their properties and uses. (jan 2010, May 2019)